

Shubh Soni

Software Engineer | Toronto, ON

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EDUCATION

Advanced Diploma, Software Engineering

Jan 2023 – Apr 2025

Conestoga College, Waterloo, ON

Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, System Design

TECHNICAL SKILLS

Languages: Java, JavaScript, Python, Bash, SQL

Backend: Spring Boot, Node.js, FastAPI, Flask

Frontend: React.js, React Native, Next.js, Tailwind CSS

Databases: Oracle DB, PostgreSQL, MySQL, MongoDB, Kafka, Redis

Cloud & DevOps: Azure, Kubernetes, Docker, AWS, GitHub Actions, CI/CD

APIs & Integration: REST APIs, JWT/OAuth

Design & Tools: Git, Github, Figma, Lucidchart, Jira, JUnit, Postman, Maven

PROJECTS

Memory Mesh

FastAPI, Next.js, PostgreSQL, pgvector, OpenAI API, Docker

- Engineered semantic memory layer for AI applications using FastAPI and PostgreSQL with pgvector extension, enabling context-aware conversations through vector similarity search
- Architected RESTful API with 8 endpoints handling memory CRUD operations, semantic search, and context retrieval with sub-200ms query latency
- Implemented hybrid search combining vector embeddings and metadata filtering, achieving 40% improved context relevance over baseline retrieval methods

Handwritten Text Recognition System

Flask, Google Vision API, Google Translate API, Redis

- Developed handwritten OCR web application achieving 98% text extraction accuracy by implementing Google Vision API with custom preprocessing, eliminating manual transcription for 2,000+ users
- Established multi-format document generation pipeline, enabling users to download recognized text in three formats
- Constructed protected user management system with activity logging and strengthened authentication

Emergency Health QR

Flask, Python, JavaScript, PostgreSQL, REST API, Encryption

- Created a Flask-based Emergency Health Card system with secure authentication and QR code integration for instant emergency data retrieval.
- Developed a responsive dashboard with dynamic form handling for multiple emergency contacts and medical information.
- Enhanced data security through comprehensive encryption and input validation protocols, reducing vulnerability exposure by 80%.

Kindle Library

Java, Spring Boot, SQLite, Firebase, Android Studio

- Built a full-stack Kindle Library application with a Java-based Android frontend and a Spring Boot backend API for secure data management and authentication.
- Integrated Firebase for user login and real-time synchronization, and SQLite for local offline caching to ensure reliability in low-network environments.
- Created a personalized recommendation engine that improved user engagement by 70%, and implemented advanced search filters by genre, author, and price with real-time results processing.

EXPERIENCE

Software Developer

Freelance

May 2023 – Present

- Eliminated manual stock discrepancies for a retail client by building a real-time inventory management system in Python and MySQL, processing 500+ daily transactions
- Enabled clients to self-manage 100+ product listings by building a Next.js/TypeScript storefront with Sanity headless CMS, styled in Tailwind and deployed on Vercel
- Cut order-processing turnaround by 45% by designing and automating a workflow that connected third-party APIs into a client's existing operations

Software Engineering Academic Tutor

Conestoga College

May 2023 – Dec 2023

- Delivered one-on-one tutoring sessions in Data Structures, Algorithms, and version control to facilitate project success
- Led collaborative sessions using Git and CI/CD pipelines for version control and testing
- Introduced pair programming techniques that fostered collaborative coding skills and teamwork